

POZNAN KRZESINY, Poland

WMO: 123260

Lat: 52.332N

Lon: 16.966E

Elev: 265

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 6.6	(c) 12.3	(d) 1.5	(e) 6.0	(f) 8.8	(g) 6.8	(h) 7.9	(i) 15.0	(j) 26.0	(k) 40.1	(l) 23.2	(m) 39.2	(n) 5.3	(o) 100	(p) 0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	87.8	67.5	84.0	66.0	80.8	64.8	70.1	82.5	68.5	79.5	66.9	76.7	9.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.0	73.7	64.7	92.6	72.2	63.3	87.8	71.1	34.2	83.0	32.8	79.5	31.6	76.9	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 22.0	(b) 18.9	(c) 16.7	(e) 2.1	(f) 93.1	(g) 8.5	(h) 3.3	(i) -4.0	(j) 95.5	(k) -9.0	(l) 97.4	(m) -13.8	(n) 99.2	(o) -20.0	(p) 101.6		
			DB	2.1	93.1	8.5	3.3	-4.0	95.5	-9.0	97.4	-13.8	99.2	-20.0	101.6	
			WB	1.3	72.9	7.5	1.9	-4.1	74.3	-8.5	75.4	-12.7	76.5	-18.2	77.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	29.8	32.2	39.1	49.4	56.9	63.8	67.5	66.8	58.1	48.6	40.5	34.3
	DBStd	14.95	9.78	8.94	8.21	7.18	6.91	5.82	5.67	5.22	5.72	6.93	6.19	8.10
	HDD50	2477	626	500	346	97	17	0	0	0	4	111	288	488
	HDD65	6131	1091	919	804	470	264	90	38	41	218	508	735	953
	CDD50	2112	0	0	7	78	232	414	542	521	248	68	2	0
	CDD65	293	0	0	0	1	14	54	115	97	12	0	0	0
	CDH74	2958	0	0	0	30	210	580	1113	881	143	1	0	0
	CDH80	861	0	0	0	5	39	157	377	252	31	0	0	0
(14) Wind	WSAvg	8.0	8.9	8.7	8.9	8.5	7.7	7.7	7.6	7.2	7.3	7.6	7.9	8.7
(15) Precipitation	PrecAvg	21.5	1.4	1.2	1.5	1.2	2.1	2.4	3.3	2.3	1.7	1.4	1.4	1.5
	PrecMax	28.4	3.0	2.5	3.2	3.1	4.2	4.5	7.4	5.8	4.3	3.7	3.9	3.9
	PrecMin	14.3	0.2	0.2	0.4	0.3	0.4	0.2	0.6	0.4	0.3	0.2	0.0	0.3
	PrecStd	3.3	0.7	0.6	0.7	0.7	1.0	1.2	1.8	1.1	0.9	0.9	0.7	0.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.7	54.6	64.8	77.7	84.2	89.8	91.8	91.8	84.2	72.1	58.4	53.2
		MCWB	48.0	48.1	52.6	59.0	65.1	69.4	69.1	70.5	66.3	61.0	53.3	48.8
	2%	DB	48.0	50.0	59.1	72.1	80.0	84.3	88.3	86.2	77.8	66.8	54.5	49.5
		MCWB	44.5	45.0	48.9	56.5	63.1	66.5	67.1	67.5	63.4	58.1	50.8	46.6
	5%	DB	45.0	46.7	55.1	67.5	75.2	80.4	84.4	82.3	73.9	63.0	51.8	47.1
		MCWB	42.2	42.3	47.6	54.1	60.8	63.7	65.8	66.4	61.2	56.7	48.6	44.6
	10%	DB	42.3	43.7	51.5	63.5	71.1	76.7	80.5	79.0	69.8	59.4	49.7	44.5
		MCWB	39.9	40.3	45.3	51.8	58.3	61.9	65.1	64.8	59.4	54.7	47.3	42.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	49.4	53.8	60.9	67.5	71.9	71.9	72.2	67.8	62.1	54.0	49.5
		MCDB	51.9	52.8	62.3	74.4	78.9	84.6	85.0	85.8	81.0	69.4	57.2	51.5
	2%	WB	45.0	45.6	50.5	58.1	64.8	68.7	70.0	70.0	64.9	59.3	51.2	47.0
		MCDB	47.8	49.4	56.9	69.3	76.1	81.2	81.9	82.0	74.8	65.1	53.6	49.0
	5%	WB	42.4	42.7	48.3	55.4	62.3	66.2	68.3	68.3	62.7	57.2	49.4	44.6
		MCDB	44.8	46.1	53.9	65.6	72.6	76.4	79.6	78.0	70.7	62.2	51.5	46.8
	10%	WB	39.9	40.3	46.0	52.9	59.9	64.2	66.8	66.6	60.7	55.1	47.3	42.5
		MCDB	42.2	43.6	51.0	62.0	68.9	72.0	76.9	74.6	67.7	59.3	49.4	44.4
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	10.6	14.4	19.4	19.3	19.0	18.3	19.2	17.7	14.5	9.7	8.2
		MCDBR	9.5	14.2	21.7	26.9	26.5	26.0	26.0	25.9	25.2	20.6	13.7	9.6
		MCWBR	7.7	10.4	13.5	14.3	12.4	11.2	9.1	10.9	12.3	12.8	10.4	8.0
	5% WB	MCDBR	9.4	14.0	19.6	24.6	23.6	22.0	21.7	21.2	21.5	18.6	13.6	9.2
		MCWBR	8.0	10.7	12.6	13.5	12.1	11.1	9.5	10.1	11.5	12.2	10.9	8.0
(40) Clear-Sky Solar Irradiance	taub		0.316	0.343	0.378	0.421	0.416	0.412	0.438	0.431	0.390	0.368	0.345	0.311
	taud		2.300	2.292	2.233	2.167	2.231	2.272	2.215	2.242	2.330	2.362	2.349	2.323
	Ebn at Noon		212	241	254	257	265	267	257	252	249	229	199	192
	Edh at Noon		20	27	35	42	42	40	42	39	32	25	20	17
(44) All-Sky Solar Radiation	RadAvg		216	431	795	1319	1620	1699	1585	1399	1010	557	246	155
	RadStd		33	74	104	171	183	151	193	127	126	77	36	18

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (29 neighbors)	+0.77	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+106	N/A

Nomenclature: See separate page